

Two Counterexamples for Normalized Separating Hyperplanes

Negative answers to both open questions on page 7 of arXiv:1508.05249

Automated mathematics run `fa_banach.001`, lane 10

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Status. Candidate counterexample, likely valid, pending human mathematical review. Both conclusions below are proved analytically; the included code is supplementary verification only.

1 Source questions and answer

Steinwart’s paper [1] studies a continuous functional $\Gamma : B \rightarrow \mathbb{R}$ on a convex subset of a normed space and the unique normalized functionals separating its level sets. On PDF page 7 it asks:

“Finally, whether $r \mapsto h_r$ is actually measurable with respect to $L_\infty(\mu)$ remains an open question.”

After proving that the weak level-set continuity condition G5 is automatic in finite dimensions, the paper also asks:

“Whether this is true in more general settings remains an open question.”

The source passages are reproduced below at full text width.

$\tilde{h} : I \times \Omega \rightarrow \mathbb{R}$ of this family that is $\hat{\mathcal{B}}(I) \otimes \mathcal{A}$ -measurable, and in turn this measurability can be used to make the second part of Osband’s principle work, see [19] for a more elementary but also technically more involved approach. Moreover, our normalization in $(L_1(\mu))' = L_\infty(\mu)$ means that we have $\|h_r\|_\infty = 1$ for all $r \in I$, and the latter is the additional information provided by Corollary 3.2 when compared to Theorem 3.1. Finally, whether $r \mapsto h_r$ is actually measurable with respect to $L_\infty(\mu)$ remains an open question.

Figure 1: The L^∞ -measurability question, source PDF page 7.

We answer both questions negatively, inside the motivating model of bounded probability densities. The first example has a continuous, strictly quasi-monotone property but fails G5. The second has normalized representatives which are continuous in every finite L^p but are not even norm-Borel measurable in L^∞ .

2 Framework from the source paper

Let $(\Omega, \mathcal{A}, \mu)$ be a probability space,

$$E = L^1(\mu), \quad B = \{p \in L^\infty(\mu) : p \geq 0, \int p \, d\mu = 1\}.$$

Corollary 4.2. *Let **B1**, **B2***, **B3**, **B4**, **G1***, **G2** and be satisfied, and E be finite dimensional. Then, the map $Z : I \rightarrow E'$ defined by*

$$Z(r) := \hat{z}'_r,$$

where $\hat{z}'_r \in E'$ are the unique functionals obtained in Theorem 2.4, is norm continuous.

Note that for finite dimensional spaces E , condition **B4** reduces to $H = E$, that is $E = \text{span } B$. Moreover, in the case of the example discussed around (4) a finite dimension of E means that Ω is finite.

Finally, note that condition **G5** does not appear in Corollary 4.2. Since **B6** is automatically satisfied for finite dimensional spaces, we thus conclude by Theorem 4.1 that **G5** always holds in this setting. Whether this is true in more general settings remains an open question.

Figure 2: The G5 question and its finite-dimensional context, source PDF page 7.

This is exactly the bounded-density set in equation (4) of [1]. We use its standard facts, spelling them out because they are essential to the counterexamples.

Take $p_\star = \mathbf{1}$. Then $H = \text{span } B = L^\infty(\mu)$ and

$$F = \text{span}(-p_\star + B) = \{f \in L^\infty(\mu) : \int f d\mu = 0\}.$$

With $\|\cdot\|_F = \|\cdot\|_\infty$, the F -norm dominates the L^1 -norm and 0 is an interior point of $-p_\star + B$ in F . Thus **B1** and **B2*** hold. Positive and negative parts give **B3** (with $K = 1$ for the L^1 norm), and density of L^∞ in L^1 gives **B4**.

For a family $\phi_r \in L^\infty(\mu)$, suppose that for every $p \in B$ the scalar function

$$F_p(r) := \int \phi_r p d\mu$$

is continuous, strictly monotone, and has a unique zero. Denote that zero by $\Gamma(p)$. Then

$$\{\Gamma = r\} = \{p \in B : F_p(r) = 0\}$$

is convex. Moreover, if $\Gamma(p) = u < v = \Gamma(q)$, every nontrivial convex mixture of p and q has its zero strictly between u and v . Therefore Γ is quasi-monotone and is strictly quasi-monotone on the interior of its image. Once continuity is verified, Theorem 2.4 of [1] shows that **G1*** and **G2** hold and identifies the unique normalized separating functionals.

3 Proof intuition

Normalization is the only source of pathology in the first example. We make unnormalized separators z_r converge weak-star as $r \downarrow r_0$, while a “bump escaping to infinity” keeps $\|z_r\|_\infty$ large for every $r > r_0$. At r_0 that bump disappears pointwise and the norm drops. Dividing by the norm therefore changes the weak-star limit by a nontrivial scalar. The source paper’s equivalence between **G5** and weak-star continuity then gives failure of **G5**.

For measurability, moving threshold functions are ideal: changing the threshold on a short interval is small in every finite L^p , but its L^∞ size is unchanged. After normalization, any two distinct thresholds remain more than one apart in L^∞ . Open unions of disjoint balls can consequently encode arbitrary subsets of the parameter interval, including nonmeasurable ones.

4 Main result

Theorem 1 (Full negative answers). *The following statements hold.*

- (A) *There exist a probability space, $E = L^1(\mu)$, the bounded-density set B above, and an L^1 -continuous functional $\Gamma : B \rightarrow \mathbb{R}$ satisfying B1, B2*, B3, B4, B6, G1*, and G2, for which G5 fails.*
- (B) *There exist an atomless probability space and such a functional Γ for which the unique normalized representing curve $r \mapsto h_r \in L^\infty(\mu)$ is not measurable from the completed Lebesgue sigma-algebra on I to the norm-Borel sigma-algebra of $L^\infty(\mu)$. Nevertheless, it is continuous as an L^p -valued map for every $1 \leq p < \infty$. The auxiliary-space hypotheses of Corollary 3.2 in [1] hold with $E_0 = L^2(\mu)$.*

5 A continuous property for which G5 fails

Let $\Omega = \mathbb{N}$, let $\mu(\{k\}) = 2^{-k}$, and use the bounded-density model above. Then $E = L^1(\mu)$ is an infinite-dimensional separable Banach space, so B6 holds. Set

$$r_0 = \frac{1}{2}, \quad \delta = \frac{1}{16}, \quad \varepsilon = \frac{1}{4},$$

and

$$a_1 = r_0 + \delta, \quad a_2 = r_0 - \delta, \quad a_k = r_0 \quad (k \geq 3).$$

For $r \in \mathbb{R}$ and $k \in \mathbb{N}$, define

$$q_r(k) = \begin{cases} 0, & r \leq r_0, \\ \min\{1, (r - r_0)k\}, & r > r_0, \end{cases} \quad \phi_r(k) = r - a_k + \varepsilon q_r(k).$$

For $p \in B$, put $F_p(r) = \int \phi_r p \, d\mu$.

5.1 The root functional satisfies the source hypotheses

For fixed k , $r \mapsto q_r(k)$ is continuous and nondecreasing. Dominated convergence therefore gives continuity of F_p , and for $s > r$,

$$F_p(s) - F_p(r) = (s - r) + \varepsilon \int (q_s - q_r) p \, d\mu \geq s - r.$$

Thus F_p is strictly increasing. At $r_0 - \delta$, every coordinate of ϕ_r is nonpositive; at $r_0 + \delta$, every coordinate is nonnegative and the q_r term is strictly positive. Hence

$$F_p(r_0 - \delta) \leq 0 < F_p(r_0 + \delta),$$

so F_p has a unique zero in $[r_0 - \delta, r_0 + \delta)$. Define $\Gamma(p)$ to be this zero.

The functional is L^1 -continuous. Indeed, if $u = \Gamma(p) < v = \Gamma(q)$, then

$$v - u \leq F_p(v) - F_p(u) = F_p(v) = \int \phi_v(p - q) \, d\mu \leq \|\phi_v\|_\infty \|p - q\|_1.$$

On the root interval $\|\phi_v\|_\infty \leq 2\delta + \varepsilon$, giving a uniform Lipschitz bound (and the case $v < u$ is symmetric).

Let $p^{(j)} = \mu(\{j\})^{-1} \mathbf{1}_{\{j\}}$. Direct calculation gives

$$\Gamma(p^{(2)}) = r_0 - \delta, \quad \Gamma(p^{(1)}) = r_0 + \frac{\delta}{1 + \varepsilon}.$$

Since B is convex and Γ is continuous, the interior I of $\Gamma(B)$ contains r_0 and a right-neighborhood of r_0 .

Level sets are affine sections because $\{\Gamma = r\} = \{p : F_p(r) = 0\}$. If $\Gamma(p) = u < v = \Gamma(q)$ and $0 < \theta < 1$, then, for $m = (1 - \theta)p + \theta q$,

$$F_m(u) = \theta F_q(u) < 0, \quad F_m(v) = (1 - \theta)F_p(v) > 0.$$

Thus $u < \Gamma(m) < v$. This proves strict quasi-monotonicity on B , hence in particular on $\Gamma^{-1}(I)$. Theorem 2.4 of [1], together with the already verified B-assumptions and continuity, yields G1* and G2.

5.2 Normalization breaks weak-star continuity

Because F_p is increasing, the correctly oriented separator is $z_r = -\phi_r$: one has $\Gamma(p) < r$ exactly when $\langle z_r, p \rangle < 0$. The unique normalized separator is therefore

$$\widehat{z}_r = \frac{z_r}{\|z_r\|_\infty} \in (L^1(\mu))' = L^\infty(\mu).$$

At $r = r_0$,

$$z_{r_0} = (\delta, -\delta, 0, 0, \dots), \quad \|z_{r_0}\|_\infty = \delta.$$

Now let $t > 0$ be small. On sufficiently large coordinates k we have $tk \geq 1$, hence $|z_{r_0+t}(k)| = \varepsilon + t$. The first two coordinates are

$$\phi_{r_0+t}(1) = -\delta + (1 + \varepsilon)t, \quad \phi_{r_0+t}(2) = \delta + (1 + 2\varepsilon)t,$$

and all remaining coordinates have absolute value at most $\varepsilon + t$. For all sufficiently small t these observations give the exact norm

$$\|z_{r_0+t}\|_\infty = \varepsilon + t.$$

For each fixed k , $q_{r_0+t}(k) \rightarrow 0$ as $t \downarrow 0$. The family is uniformly bounded, so dominated convergence against every $x \in L^1(\mu)$ gives

$$z_{r_0+t} \xrightarrow{w^*} z_{r_0}.$$

Consequently,

$$\widehat{z}_{r_0+t} \xrightarrow{w^*} \frac{z_{r_0}}{\varepsilon} = \frac{\delta}{\varepsilon} \widehat{z}_{r_0} = \frac{1}{4} \widehat{z}_{r_0} \neq \widehat{z}_{r_0}.$$

Choose, for example, $t_n = \delta/(4n)$. The preceding image calculation ensures $r_0 + t_n \in I$. Thus the normalized separator curve is not weak-star continuous at r_0 . All hypotheses of Theorem 4.1 in [1] are satisfied, and that theorem states that G5 is equivalent to weak-star continuity of this curve. Hence G5 fails. This proves Theorem 1(A).

6 Finite- L^p continuity but failure of L^∞ measurability

Let $\Omega = (0, 1)$ with Lebesgue probability measure. For $0 \leq r \leq 1$ set

$$\phi_r(t) = 1 - 2\mathbf{1}_{(0,r]}(t) - r, \quad F_p(r) = \int_0^1 \phi_r(t)p(t) dt = 1 - r - 2 \int_0^r p(t) dt.$$

For every $p \in B$, F_p is continuous, $F_p(0) = 1$, $F_p(1) = -2$, and, for $s > r$,

$$F_p(s) - F_p(r) = -(s - r) - 2 \int_r^s p(t) dt \leq -(s - r).$$

It therefore has a unique zero; again call it $\Gamma(p)$.

Every $r \in (0, 1)$ occurs as a zero. Indeed, the bounded density

$$p_r(t) = \begin{cases} \frac{1-r}{2r}, & 0 < t < r, \\ \frac{1+r}{2(1-r)}, & r < t < 1 \end{cases}$$

has total mass one and $\int_0^r p_r = (1-r)/2$, hence $\Gamma(p_r) = r$. Thus $\Gamma(B) = (0, 1)$ and $I = (0, 1)$.

If $u = \Gamma(p) < v = \Gamma(q)$, monotonicity gives

$$v - u \leq |F_p(v) - F_q(v)| \leq \|\phi_v\|_\infty \|p - q\|_1 < 2\|p - q\|_1.$$

So Γ is L^1 -Lipschitz. The same two-endpoint argument used above, with inequalities reversed because F_p is decreasing, shows strict quasi-monotonicity. Hence the source paper's G1* and G2 conditions hold.

Here the separator already has the correct orientation: $z_r = \phi_r$, and

$$\|z_r\|_\infty = 1 + r, \quad h_r := \widehat{z}_r = \begin{cases} -1, & 0 < t \leq r, \\ c_r := \frac{1-r}{1+r}, & r < t < 1. \end{cases}$$

If $0 < r < s < 1$, then on $(r, s]$ the two functions take the values c_r and -1 . Elsewhere their difference is no larger, so

$$\|h_r - h_s\|_\infty = 1 + c_r = \frac{2}{1+r} > 1. \quad (1)$$

Thus the image is an uncountable set whose distinct points are separated by more than one.

For any subset $A \subset I$, let

$$U_A = \bigcup_{r \in A} B_{L^\infty}(h_r, 1/2).$$

This is open in the norm topology of L^∞ . By (1),

$$\{s \in I : h_s \in U_A\} = A.$$

Choose A outside the completed Lebesgue sigma-algebra on I . Then the inverse image of the norm-open set U_A is not measurable. Hence $r \mapsto h_r$ is not norm-Borel measurable and, a fortiori, is not Bochner measurable.

This failure is invisible in every finite L^p . For $1 \leq p < \infty$ and $r < s$, direct integration gives

$$\|h_r - h_s\|_p^p = (s - r) \left(\frac{2}{1+r} \right)^p + (1 - s)|c_r - c_s|^p \longrightarrow 0 \quad (s \rightarrow r).$$

The analogous formula from the left proves L^p -continuity on all of I .

Finally, this is within the precise auxiliary-space setting of Corollary 3.2 of [1]. Take $E_0 = L^2(0, 1)$, continuously embedded in $L^1(0, 1)$. It is a separable Hilbert space, $B \subset L^\infty \subset L^2$, and the B2–B4 verification above works with the same L^∞ dominating norm. Moreover, B is Borel in L^2 : it is the intersection of the closed positive cone and the closed integral-one hyperplane with

$$L^\infty = \bigcup_{m=1}^{\infty} \{f \in L^2 : |f| \leq m \text{ a.e.}\},$$

whose displayed pieces are L^2 -closed. Thus G4 and B5 hold, and L^1 -Lipschitz continuity implies L^2 -continuity. This proves Theorem 1(B).

7 Verification and limitations

The included script `code/check_counterexamples.py` deterministically checks 80 sampled discrete densities, 80 exact separator scales, 100 threshold levels, all pairwise L^∞ gaps on that grid, the explicit witnessing densities, and finite- L^p limits. It prints

```
PASS: both counterexample sanity suites completed
      example A: 80 random densities and 80 separator scales
      example B: 100 levels, pairwise L-infinity separation, finite-Lp limits
```

These computations are sanity checks only; no finite computation proves the theorem.

The first counterexample uses Theorems 2.4 and 4.1 of the source paper for the equivalence with G2 and G5. The second conclusion is specifically about the norm-Borel structure of L^∞ , the Banach-valued/Bochner interpretation used in the source’s Theorem 3.1 and the surrounding question. Its curve is in fact weak-star continuous, so the packet makes no claim against weak-star measurability.

8 Novelty check and review recommendation

On 11 August 2026, bounded searches covered the run registry, solution index, and attempt index, plus open web/arXiv searches for the arXiv id, exact title, author, the exact question phrases, “weak level set continuity,” “G5,” and close L^∞ -measurability variants. They found the source paper but no later paper explicitly answering either question. This is bounded negative evidence, not an exhaustive novelty claim.

The packet is recommended for high-priority human review as a candidate full counterexample. The principal audit points are the invocation of Theorem 2.4 after strict quasi-monotonicity, the norm computation in the escaping-coordinate example, and the arbitrary-open-union argument in L^∞ .

References

- [1] Ingo Steinwart, *Representation of Quasi-Monotone Functionals by Families of Separating Hyperplanes*, arXiv:1508.05249v1 [math.OC], 2015. <https://arxiv.org/abs/1508.05249>.